(i) A student states

"If $x$ and $y$ are irrational numbers, $x \neq y$, then $xy$ is also irrational."

Show, by counter example, that this statement is not always true.

**(1)**

(ii) Prove, using algebra, that for all odd integers $n$, the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

**(4)**